Photosynthesis Learning Guide

The high-yield science behind your CER test, tied to your own packets

Sources: [P1] Photosynthesis and Respiration · [P2] the simulation lab · [P3] your Mauna Loa notes

How to use this guide

- **Parts 1–4:** the core science for every **Reasoning** section. **Parts 5–6 matter most:** the lab, your own Part III data (your **Evidence**) and what controls the rate — lab Part VII is the model for the test. **Parts 7–9:** carbon cycle, humans, Keeling Curve. **Parts 10–11:** vocabulary and a self-check.
- **The other guides:** Guide 2 (CER Prep Guide) — how to write C-E-R · Guide 3 (Mock CER Questions) · Guide 4 (Cheat Sheet) · Guide 5 (Detected Errors) — all your worksheet fix-its, in detail. Codes: [P1 Q9] = Packet 1, question 9 · [P2 Part III] = the lab · [P3] = your Mauna Loa notes.

WHAT'S INSIDE

- Part 1 · The big picture: energy flows, matter cycles
- Part 2 · Photosynthesis up close
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- Part 4 · Partners: how the two connect
- Part 5 · Measuring it: the simulation lab
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- Part 7 · The carbon cycle
- Part 8 · Humans & the carbon cycle
- Part 9 · Mauna Loa & the Keeling Curve
- Part 10 · Vocabulary
- Part 11 · Quick self-check

● KEY IDEA — The 60-second version

- **Photosynthesis** (chloroplasts): $6CO_2 + 6H_2O$ + light energy → $C_6H_{12}O_6 + 6O_2$. Light energy is stored as chemical energy in glucose.
- **Cellular respiration** (mitochondria; plants AND animals; 24/7): $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy (ATP).
- Each one's products are the other's reactants: **matter cycles; energy flows** (in as light, out as heat).
- Lab: **DO difference** (net) = O_2 made − O_2 used; **O_2 production** (gross) = DO difference + 0.1939; near zero at 20–40 % light, then a steady rise — light was the **limiting factor** from 40 % to 100 %.
- **Carbon cycle:** photosynthesis removes CO_2; respiration and combustion add it. **Keeling Curve:** a yearly zig-zag on a long rise (about 313 ppm in 1958 → about 432 ppm in May 2026).

Part 1 · The big picture: energy flows, matter cycles

Every living thing needs **energy** and **matter** (atoms to build its body). **Producers** (plants, algae) make their own food by photosynthesis. **Consumers** (animals) eat other living things. **Decomposers** (fungi, bacteria, worms) break down dead things and wastes, returning matter to the air and soil.

Following the energy: from the sun to an athlete [P1 Q11–12]

In [P1 Q11–12] you traced your meal back to plants — exactly right. Figure 1 adds the two links a CER answer needs: how the energy is **stored** and how it is **released**.

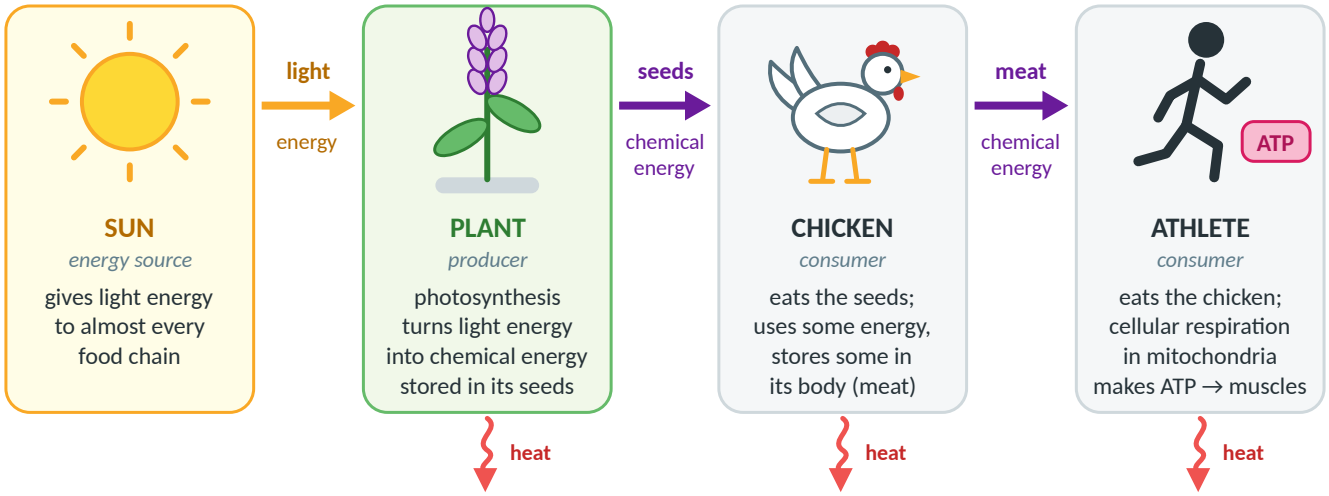


Figure 1. The energy chain from [P1 Q12]: light energy → chemical energy in food → ATP in the athlete's cells. Some energy is lost as heat at every step.

√ FIX-IT FROM YOUR WORKSHEET — [P1 Q12] — add the two missing links

Add: (1) **photosynthesis stores light energy as chemical energy in glucose/starch** in the seeds, and (2) the athlete's **cells release it by cellular respiration in their mitochondria as ATP**, which powers the muscles. For [P1 Q11], trace a second food too: pizza crust is made from wheat, a plant.

● KEY IDEA — Matter cycles; energy flows

Matter (CO_2, H_2O, glucose, O_2) is **recycled**: the same atoms go round and round [P1 Q3].

Energy is not recycled: it enters as light, is stored in food, is used as ATP and leaves as heat — so sunlight must be constantly available [P1 Q4].

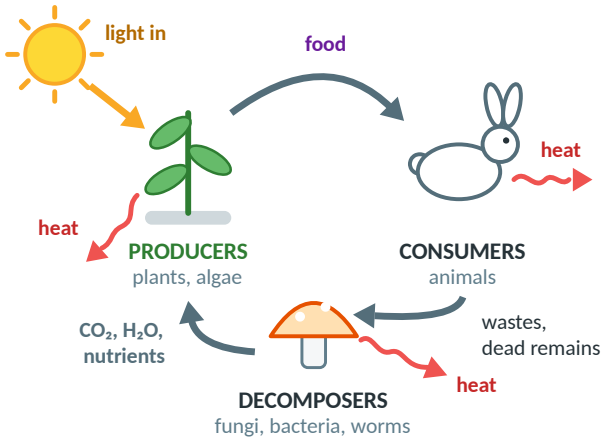


Figure 2. Matter (gray) loops; energy (yellow in, red out) flows through once.

Part 2 · Photosynthesis up close

Photosynthesis is the process in which producers use **light energy** to turn **carbon dioxide** and **water** into **glucose** (a sugar) and **oxygen**. It happens in **chloroplasts**, where the green pigment **chlorophyll** captures the light.

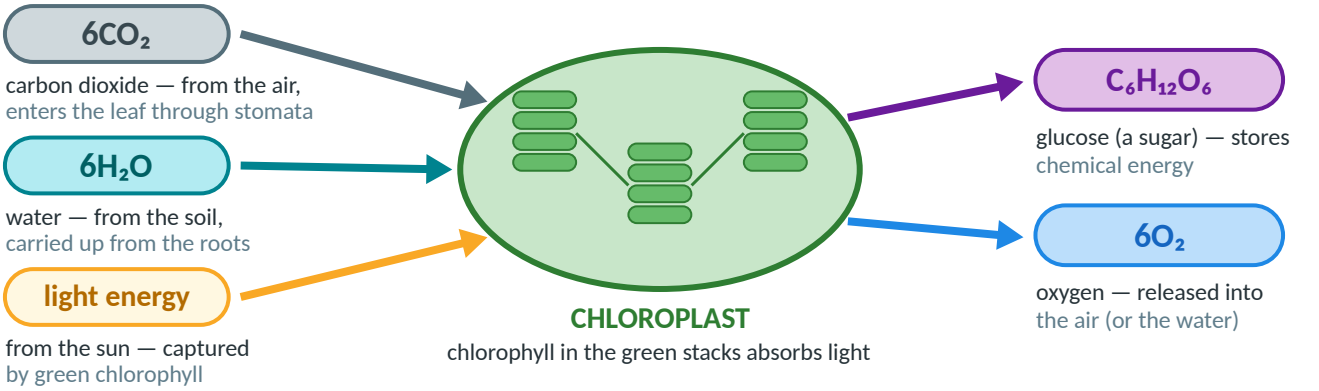
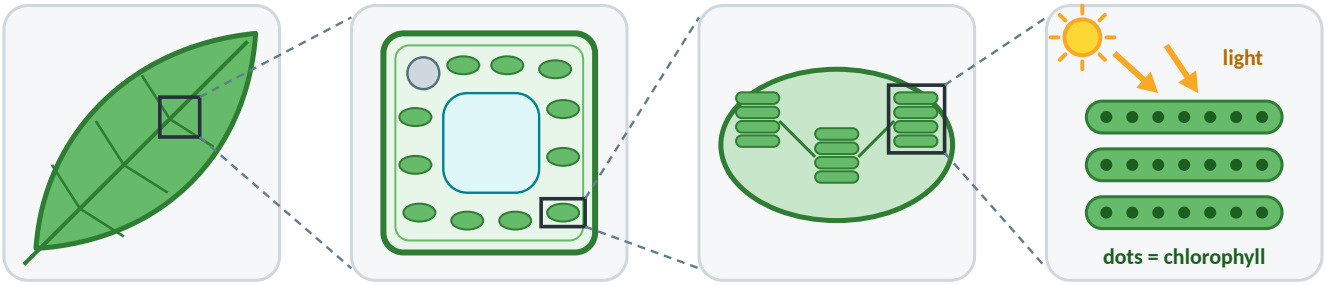


Figure 3. Photosynthesis, color-coded. Reactants go into the chloroplast; products come out. Light is the energy input.

→ REASONABLE INFERENCE (beyond the packets) — Where the inputs come from

CO_2 enters the leaf through tiny pores called **stomata**, and O_2 leaves the same way. **Water** comes up from the roots through the stem and leaf veins. (For your lab's aquatic plant, CO_2 and O_2 are dissolved in the water.)

Zooming in: leaf → cell → chloroplast → chlorophyll



each panel zooms in on the boxed part of the one before it

Figure 4. From leaf to chlorophyll. Leaf cells are packed with chloroplasts [P1 Q5].

The energy change: light → chemical

Photosynthesis **stores** energy: light energy becomes **chemical energy** in the bonds of glucose. The plant uses some glucose for its own respiration, stores some as **starch** (seeds, roots, fruit) and builds new cells with the rest.

> **→ REASONABLE INFERENCE (beyond the packets) — Enrichment: where does the O_2 come from?**
> From **splitting water** (H_2O) with light energy — not from CO_2, whose carbon ends up in glucose.

Part 3 · Cellular respiration up close

Cellular respiration is the process in which cells **break down glucose using oxygen** to release its stored energy as **ATP**; carbon dioxide and water are released. It happens in the **mitochondria** of plant AND animal cells.

ATP — the cell's energy currency. Glucose holds a lot of energy, but cells can't "spend" it directly. Cellular respiration transfers its energy into many small, usable packets called **ATP** — like changing a big bill into coins.

Every job a cell does is paid for with ATP: muscles contracting, building new cells, moving materials, sending nerve signals — and, in plants, growing. Some energy is always lost as **heat** along the way.

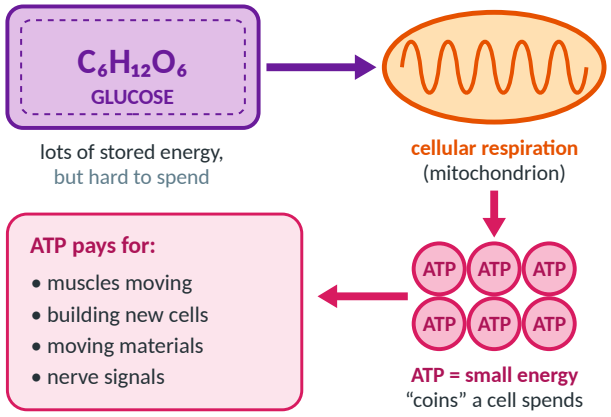
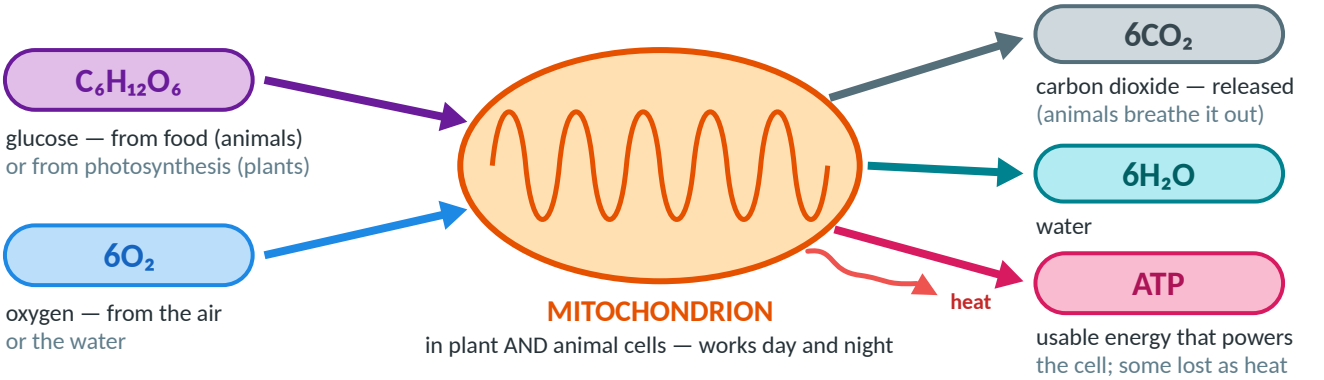


Figure 5. Respiration turns stored energy (glucose) into spendable energy (ATP).

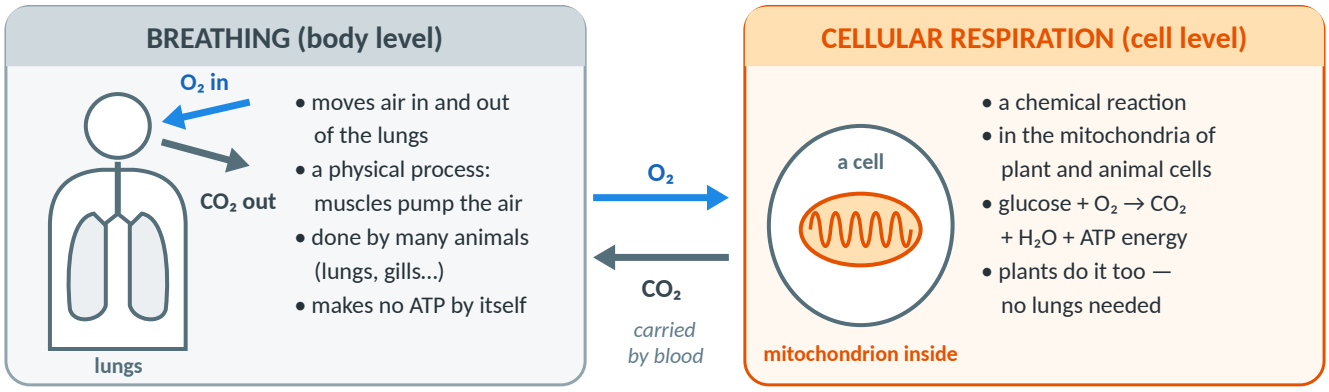


REACTANTS (go in) → PRODUCTS (come out)

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy (ATP)}$$

Figure 6. Cellular respiration, color-coded. Glucose and O_2 go into the mitochondrion; CO_2, H_2O and ATP come out, and some energy is lost as heat.

Respiration ≠ breathing



Link: breathing brings in the O_2 and gets rid of the CO_2; cellular respiration is the reaction that uses the O_2 to make ATP.

Figure 7. Breathing and cellular respiration are different things that work together.

! WATCH OUT — Two classic mistakes

- **"Respiration = breathing."** Breathing moves gases in and out of the lungs; **cellular respiration** is a chemical reaction in the mitochondria of cells.
- **Forgetting that plants respire.** Plants respire **24/7**; photosynthesis needs light. By day a plant usually makes more O_2 than it uses; at night it only respires, using O_2 and releasing CO_2.

Part 4 · Partners: how the two connect

The two processes are partners. Here they are side by side; Figure 8 redraws Model 1 [P1 Model 1] to show how they form a loop — what one makes, the other uses.

	Photosynthesis	Cellular respiration
Organelle	chloroplast	mitochondrion
Reactants (in)	$6CO_2 + 6H_2O$ + light energy	$C_6H_{12}O_6 + 6O_2$
Products (out)	$C_6H_{12}O_6 + 6O_2$	$6CO_2 + 6H_2O$ + energy (ATP)
Energy	**stores** it: light → chemical energy in glucose	**releases** it: glucose → ATP (+ heat)
Who does it	producers: plants, algae	almost all living things: plants, animals, fungi…
When	only when there is light	all the time, day and night
Effect on air/water	removes CO_2, adds O_2	removes O_2, adds CO_2

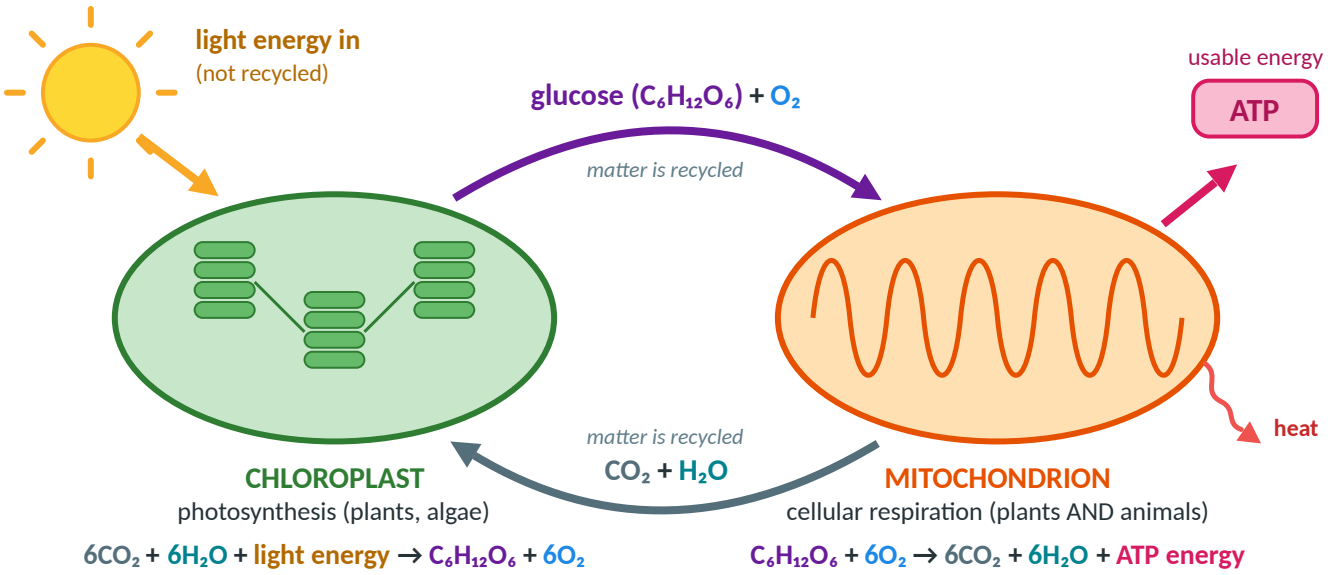


Figure 8. Model 1, redrawn. Glucose and O_2 from the chloroplast feed the mitochondrion; CO_2 and H_2O come back. Light comes in; ATP and heat go out.

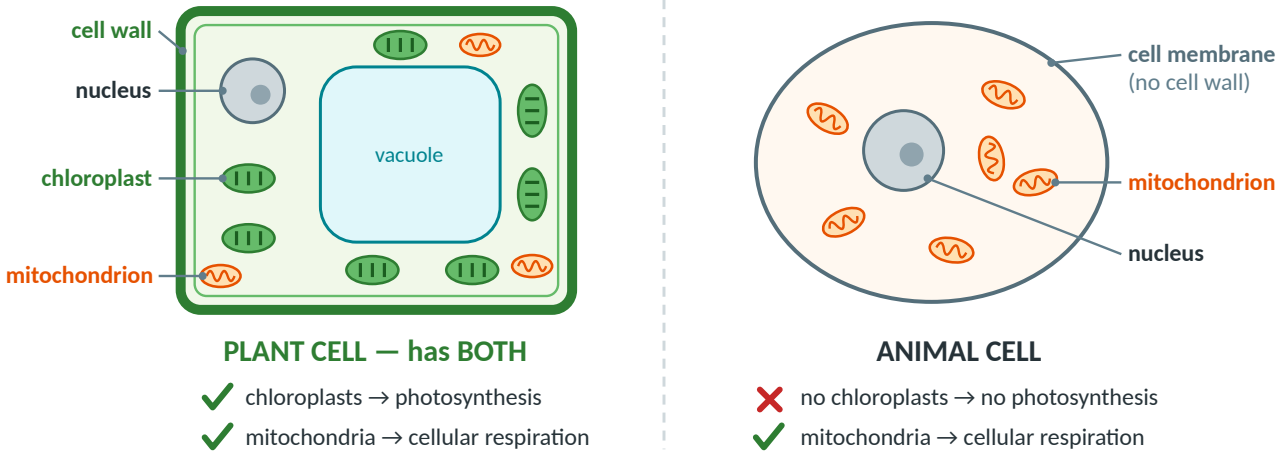


Figure 9. Plant cells have chloroplasts AND mitochondria, so they do both processes. Animal cells have mitochondria but no chloroplasts [P1 Q5–8].

Plant cells need mitochondria too: glucose is *stored* energy, and **cellular respiration** turns it into usable ATP [P1 Q6]. Animals have no chloroplasts, so they get glucose by eating [P1 Q7].

√ FIX-IT FROM YOUR WORKSHEET — Quick fix-its from Packet 1 — full details in Guide 5 (Detected Errors)

- **[P1 Q9]** First half √ correct. But glucose + O_2 are *exactly* photosynthesis's products, and ATP is a **product** of respiration. End with: *"The only difference is energy: photosynthesis takes in light; respiration releases energy as ATP."*
- **[P1 Q10]** You meant **chloroplasts** — animals do have mitochondria (your Q8). Add **interdependence**: plants use CO_2 from respiration; decomposers recycle matter.
- **[P1 Q14]** Formulas and CO_2 circles √ correct; only the energy labels got swapped: **light energy goes IN to photosynthesis; ATP comes OUT of respiration.**

Part 5 · Measuring it: the simulation lab

In the lab [P2] a simulated **aquatic plant** sat in a tank. The simulation tracked **dissolved oxygen (DO)** — the O_2 gas dissolved in the water, in **mg/L** — over a **60-minute** run; you read the start and end values off its graph.

You could change the number of fish and plants, the light intensity, the light color and the temperature. Your team did **Part III**. Driving question: *How does light intensity affect oxygen production during photosynthesis?*

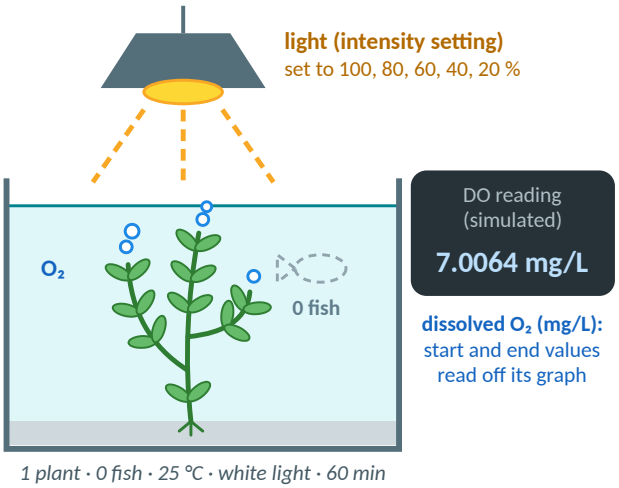


Figure 10. Your Part III setup: only the light intensity was changed.

Experimental design (CER tests ask about this)

Term	What it means	In your Part III test
Independent variable	what you changed	light intensity: 100, 80, 60, 40, 20 %
Dependent variable	what you measured	O_2 production (mg/L per hour), from DO start and end
Controlled variables	kept the same	0 fish, 1 plant, 25 °C, white light, 60-minute run
Control / baseline	factor removed	Part II dark run (0 % light): respiration only, −0.1939 mg/L per hour
Repeated trials	test run more than once	Part II: 3 trials, averaged. Part III: 1 trial per level (a limitation)

The baseline [P2 Part II]: three dark runs (0 % light) measured respiration alone. Averaging repeated trials makes the value more reliable, and the trial-to-trial spread (up to 0.0065) shows how big a difference must be before you can trust it.

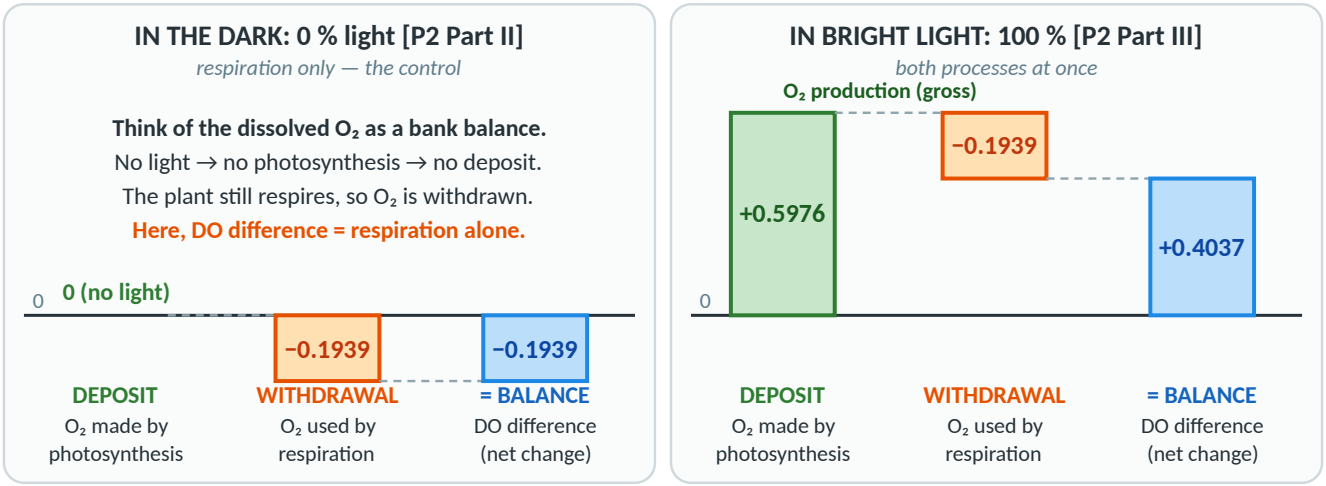
Part II trial (0 % light)	DO start (mg/L)	DO end (mg/L)	DO difference (mg/L per hour)
Trial 1	7.0735	6.8794	−0.1941
Trial 2	7.0476	6.8568	−0.1908
Trial 3	7.0735	6.8762	−0.1973
Average (your sheet)	7.0647	6.8709	**−0.1939** (exact: −0.1941)

√ FIX-IT FROM YOUR WORKSHEET — Your Part III prediction

"Light intensity decreases oxygen production" is the opposite of what you found. Predictions may be wrong, but write a testable **If… then… because…**: *"If light intensity increases, then O_2 production will increase, because light is the energy source for photosynthesis."*

DO difference vs. O_2 production: the oxygen bank account

With the light on, the plant does two things at once: photosynthesis **adds** O_2 (a deposit) and respiration **uses** O_2 (a withdrawal). The DO reading only shows the change in the balance.



O_2 production = DO difference − respiration = 0.4037 − (−0.1939) = 0.4037 **+** 0.1939 = **0.5976**

Sign trap: subtracting a negative number means ADDING it. All values are in mg/L per hour.

Figure 11. The oxygen bank account. In the dark, the only change is the respiration withdrawal. At 100 % light the DO reading shows the net change (+0.4037); the total O_2 made (0.5976) is that plus the 0.1939 used by respiration.

● KEY IDEA — Two different numbers — use the packet words

- **DO difference** (net change) = DO end − DO start = O_2 made − O_2 used. In the dark it is just respiration (−0.1939).
- **O_2 production** (gross) = all the O_2 made by photosynthesis = DO difference + 0.1939.
- **Sign trap:** the sheet says *DO difference − respiration*, and respiration is **negative**: 0.4037 − (−0.1939) = 0.4037 **+** 0.1939 = **0.5976**. Mind the time basis too: per hour (60-minute run); per minute = ÷ 60.

Your Part III data, corrected

Highlighted = corrected from your sheet [P2 Part III]. O_2 production = DO difference + 0.1939; per minute = ÷ 60.

Light (%)	DO start (mg/L)	DO end (mg/L)	DO difference (net, mg/L per hour)	O_2 production (gross, mg/L per hour)	O_2 production (mg/L per minute)
100	7.0064	7.4101	+0.4037	0.5976	**0.0100** (not 0.001)
80	7.038	7.2464	+0.2084	0.4023	0.0067
60	7.0587	7.0633	+0.0046	0.1985	0.0033
40	7.0009	6.8072	−0.1937	**0.0002** (not 0.0102)	**≈ 0** (0.000003)
20	7.0677	6.8749	−0.1928	0.0011	0.00002

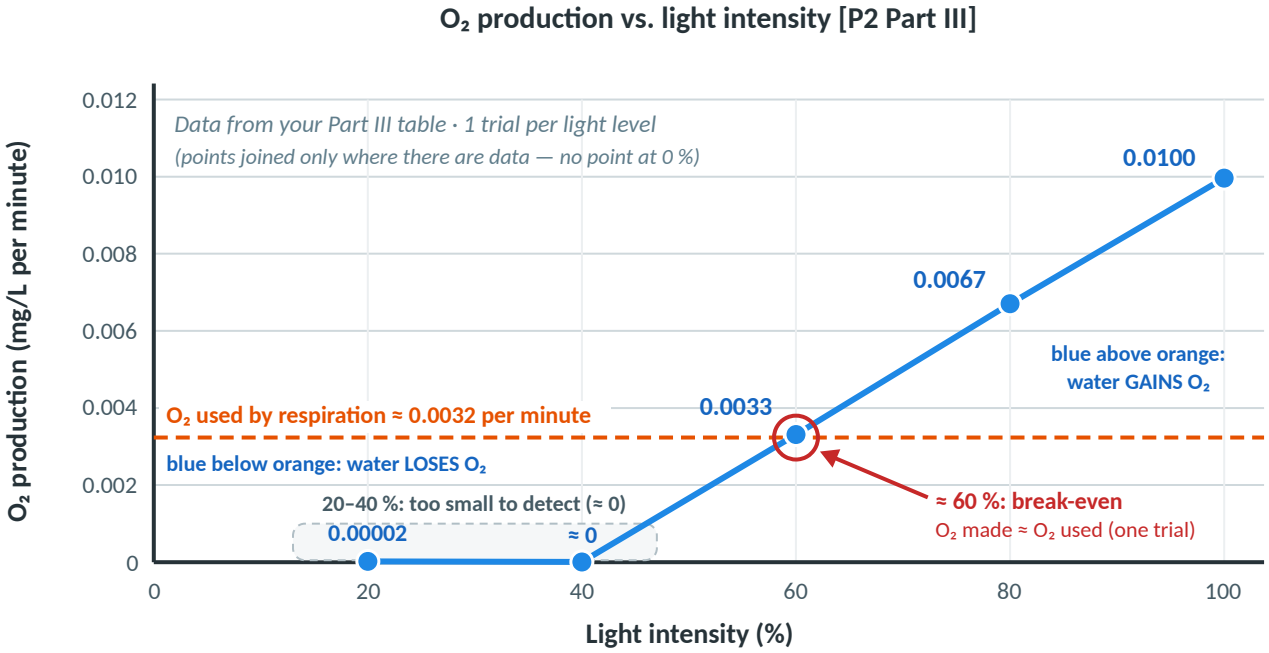


Figure 12. O_2 production per minute vs. light intensity, from your corrected table. The orange line is the O_2 used by respiration (0.1939 ÷ 60); where the blue line crosses it (near 60 %) the plant breaks even.

√ FIX-IT FROM YOUR WORKSHEET — Your Part III graph [P2 Part III Q7]

√ Axis titles with units, and 80 % and 100 % plotted correctly. To fix: your line **dips below zero near 60 %**, but your table says +0.0033 (every value is positive). Graph rules: independent variable on x, dependent on y, **even scales**, units, **mark the points**, a title, x-axis at 0, and **don't draw beyond the data** (no line back to 0 %).

What your data show

- **20 % and 40 %:** the DO fell about as much as in the dark (−0.1928 and −0.1937 vs. −0.1939), so photosynthesis was **too small to detect in these data** (0.0011 and 0.0002 are smaller than the trial-to-trial spread of about 0.003–0.006). Real plants do photosynthesize in dim light; this is just what *these* data show.
- **Near 60 % (one trial):** DO difference ≈ 0 (+0.0046): O_2 made ≈ O_2 used — a **break-even point** (the *light compensation point*; inference). Above it the water gains O_2; below it, it loses O_2.
- **From 40 % up,** each +20 % of light added about **0.20 mg/L per hour** (0.0002 → 0.1985 → 0.4023 → 0.5976): a steady straight-line rise with **no leveling off**, so light was the **limiting factor** from 40 % to 100 %. Above 100 % is outside the data — we can't say.
- **Hidden assumption (great for a Level-4 answer):** respiration can only be measured in the dark, so the worksheet assumes the plant respires at the **same rate in light as in dark**. (The tiny averaging slip, −0.1939 vs. −0.1941, changes nothing; this guide uses your −0.1939.)

Model CER — How does light intensity affect O_2 production? [P2 Part III] · see Guide 2 (CER Prep Guide)	
CLAIM answers the question	As light intensity increases, the plant's O_2 production increases: it stayed near zero at 20–40 %, then rose steadily from 40 % to 100 %.
EVIDENCE specific data / facts	• O_2 production was only 0.0011 and 0.0002 mg/L per hour at 20 % and 40 % light, then rose to 0.1985 at 60 %, 0.4023 at 80 % and 0.5976 at 100 %. • At 20 % and 40 % the DO fell by 0.1928 and 0.1937 mg/L per hour — about the same as in the dark (−0.1939).
REASONING the science "why"	Light is the energy source for photosynthesis. More light means chlorophyll in the chloroplasts absorbs more light energy, so photosynthesis runs faster and releases more O_2, one of its products. Cellular respiration uses O_2 all the time, so in dim light (20–40 %) it used O_2 much faster than photosynthesis made it, and the DO fell by about as much as in the dark. Because O_2 production never leveled off, light was the limiting factor from 40 % to 100 %.

Part 6 · What controls the rate: light, color, temperature, CO_2

A **limiting factor** is whatever is in shortest supply and holds the rate back — like the narrowest part of a pipe. The classic three for photosynthesis are **light intensity, temperature and CO_2 concentration**; light **color** matters too.

Light intensity (your team, Part III): more light → more light energy absorbed by chlorophyll → faster photosynthesis → more O_2. Your O_2 production stayed near zero at 20–40 %, then rose steadily from 40 % to 100 % with no leveling off, so light was the limiting factor from 40 % to 100 %. With even more light, another factor would eventually limit it *(inference)*. Class best: **100 %** [P2 Part VI].

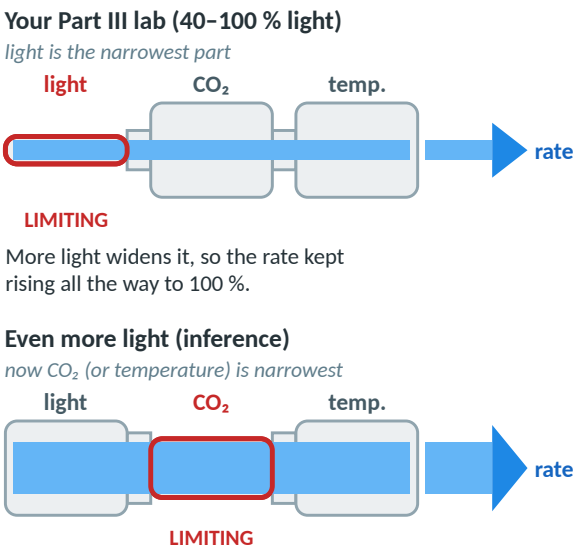


Figure 13. The limiting-factor idea: the narrowest section sets the flow.

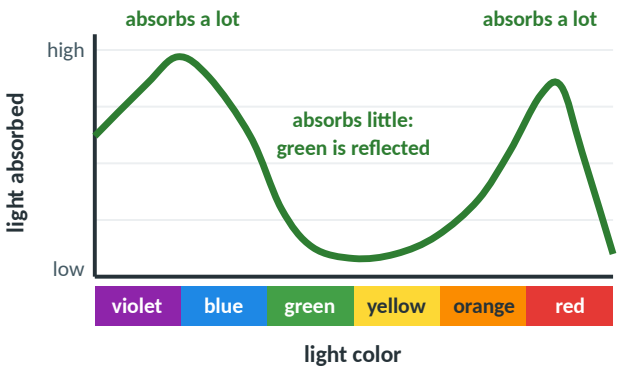
→ REASONABLE INFERENCE (beyond the packets) — CO_2 and going further

CO_2 is a reactant: more of it (e.g., baking-soda water) speeds up photosynthesis until another factor limits it — then the curve **levels off**. **Other ways to measure the rate:** count the O_2 bubbles from *Elodea*, or time how fast leaf disks float up. **Where does a plant's mass come from?** Mostly from CO_2 in the air built into glucose, not from the soil.

Light color (Part IV, another team)

Class result: **white light** gave the most O_2 [P2 Part VI]. Your reason ("white contains the most colors") is on the right track — sharpen it:

- Chlorophyll absorbs mostly **red and blue/violet** light and **reflects green** — that is why leaves look green.
- White light contains red and blue (and every other color), so the plant gets more light energy it can use.
- Among single colors, red and blue should work best and green worst.



illustrative shape (inference), not class data

Figure 14. Chlorophyll absorbs blue/violet and red, and reflects green (illustrative shape; inference).

Temperature (Part V, another team)

The class's **best tested temperature was 20 °C** [P2 Part VI]. Only 15, 20, 25, 30 and 35 °C were tested, so the true optimum is somewhere between 15 and 25 °C. Say "**the class's results** showed…", not "plants are always best at 20 °C".

Too cold: molecules move and collide more slowly, so the enzyme-driven reactions slow down. **Warmer than the optimum:** O_2 production falls because enzymes work less efficiently; **respiration also speeds up and uses more O_2**, so the net gain (the DO difference) falls even faster.

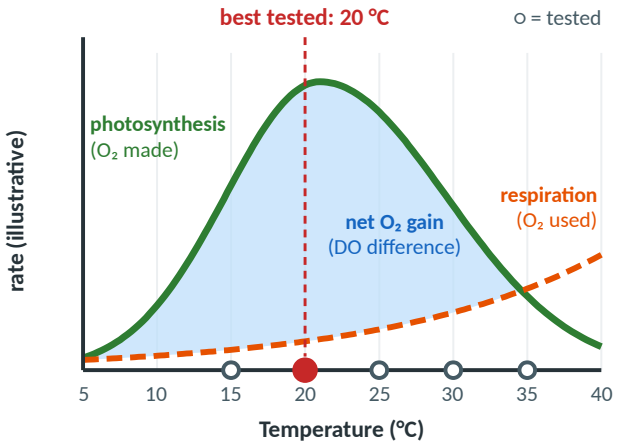


Figure 15. How temperature can affect the rates (illustrative shapes, not class data).

> **√ FIX-IT FROM YOUR WORKSHEET — "At higher temps (>20 °C), enzymes are denatured" [P2 Part VI]**
> Your cold-side reason is √ correct. But enzymes do **not** denature just above 20 °C — above the optimum they simply work less efficiently while respiration speeds up. Real denaturing of most plant enzymes needs about **40 °C or more**.

What is productivity? [P2 Part VI Q2]

√ FIX-IT FROM YOUR WORKSHEET — Unanswered on your sheet — here is the idea

Productivity = the rate at which producers **make** organic compounds (food such as glucose); the sheet's "release" wording is garbled. **Why DO?** O_2 is released in a fixed ratio as glucose is made (6 O_2 per glucose) and is easy to measure in water, so DO is a stand-in (proxy) for how fast food is made. **More than oxygen?** Yes — it really means making organic compounds (glucose → starch → new plant material); O_2 is just the easiest product to measure. The DO difference measures **net** productivity (after the plant's own respiration).

Putting it together: the optimal ecosystem [P2 Part VII]

Part VII — the model for your CER test — asks for the best conditions and **how many fish** the tank can support. Class results: **100 % light, white light, about 20 °C** (best tested). **The logic:** fish only respire (use O_2, release CO_2). The tank stays healthy only while the **DO difference stays ≥ 0**. One plant at 100 % white light and 25 °C gave **+0.4037 mg/L per hour** — that surplus is the O_2 "budget" for the fish.

- **Example — HYPOTHETICAL PRACTICE DATA:** if one fish used 0.10 mg/L per hour, the tank could support 4 fish (4 × 0.10 = 0.40 ≤ 0.4037, leaving +0.0037: a very thin margin — inside the trial-to-trial spread) but not 5 (0.50 > 0.4037). **Find the real number:** the +0.4037 was measured at 25 °C, so first re-measure the plant-only run at your chosen conditions; then add fish one at a time, rerun, and keep the last number with DO difference ≥ 0. More plants or better conditions raise the budget.
- **Interdependence [P1 Q10]:** the fish supply CO_2 for the plant's photosynthesis; the plant supplies O_2 for the fish.
- **Night check (inference):** at night there is no surplus — everything only respires. Over 24 hours (12 h light, 12 h dark) with the HYPOTHETICAL 0.10 mg/L per hour per fish (n = number of fish): 12(0.4037 − 0.10n) − 12(0.1939 + 0.10n) ≥ 0 → n ≤ 1. So a real tank supports far fewer fish unless you add plants.

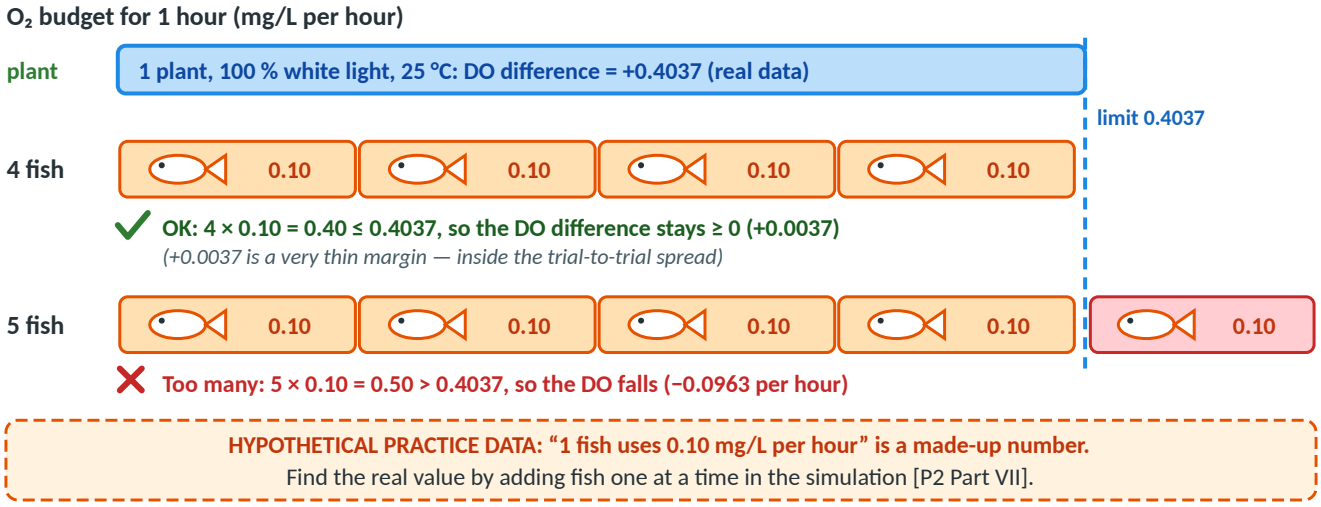


Figure 16. The O_2 budget. The plant's surplus (+0.4037, real data, measured at 25 °C) is the most the fish can use. The 0.10 mg/L per hour per fish is HYPOTHETICAL PRACTICE DATA.

! WATCH OUT — Before the test: get the real Part IV and V numbers

Your sheet is blank for Parts IV (color) and V (temperature) — correct, other teams did them. **Ask classmates for their actual data** so you can quote real numbers as evidence.

Part 7 · The carbon cycle

Model 2 [P1 Model 2] shows carbon moving between the air, living things and underground stores. The lettered arrows (A–D) move CO_2 into or out of the air.

Arrow	Process	CO_2 in the air	Who does it
A	decomposer respiration (decay)	**adds**	fungi, bacteria, worms
B	animal respiration	**adds**	animals (the rabbit)
C	plant respiration	**adds**	plants
D	photosynthesis	**removes**	plants (producers)
Combustion	burning fossil fuels	**adds**	humans: cars, factories, power plants

The other arrows (feeding, death, wastes, decay, fossil-fuel formation) move carbon between living things and the carbon stores.

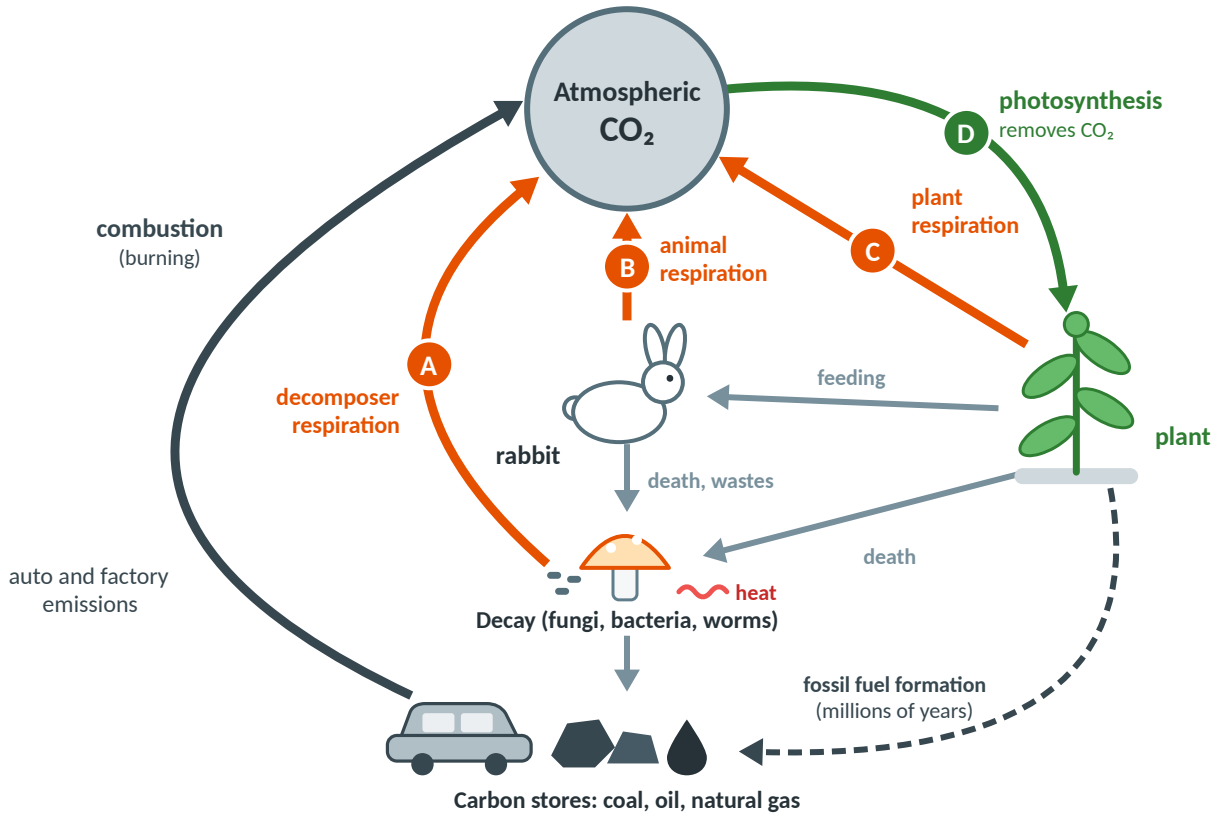


Figure 17. Model 2, redrawn. Green = photosynthesis, which removes CO_2; orange = respiration (the packet's red stars), which adds CO_2; dark gray = combustion and fossil-fuel formation; light gray = carbon moving through food, death, wastes and decay.

The natural balance [P1 Q18]

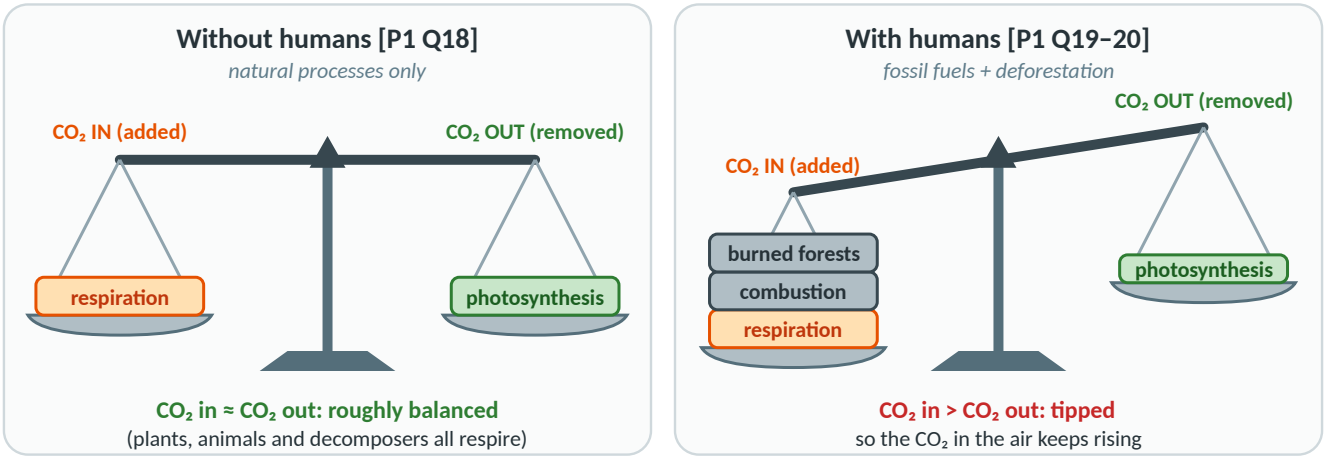


Figure 18. Your [P1 Q18] answer was strong. Left: without humans, the CO_2 added by respiration ≈ the CO_2 removed by photosynthesis, so the amount in the air stays roughly constant (an equilibrium). Right: burning fossil fuels and forests adds extra CO_2 and tips the balance [P1 Q19–20].

√ FIX-IT FROM YOUR WORKSHEET — [P1 Q13] and [P1 Q16]

Q13: green at D, red at C and A — √ correct — but you missed **B** (animal respiration). **Q16:** add **decay (decomposition)** as its own process, and swap the vague "metabolism" for *feeding/digestion* and *excretion of wastes*. Fossil-fuel formation = burial + heat + pressure over millions of years.

Part 8 · Humans & the carbon cycle

Fossil fuels (coal, oil, gas) hold *carbon* stored underground for **millions of years**. In about 200 years, humans have burned carbon that took millions of years to store, adding CO_2 far faster than nature can absorb it [P1 Q19].

Deforestation has **two** effects [P1 Q20]: fewer trees remove CO_2, **and** burning or rotting trees release their stored carbon.

The bathtub idea: CO_2 in the air is like water in a tub. It rises whenever more flows in (respiration + combustion) than drains out (photosynthesis).

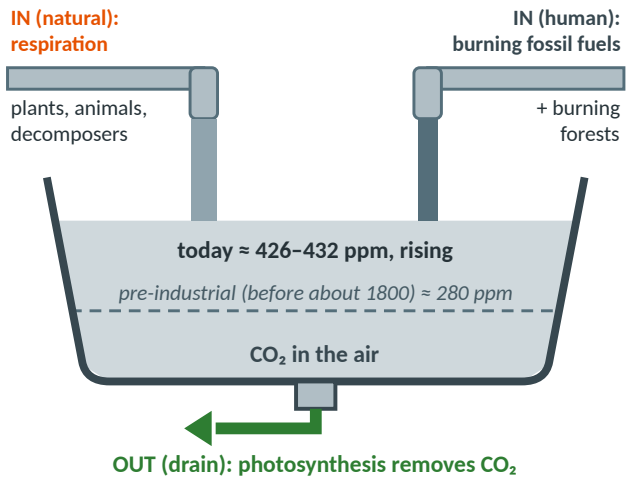


Figure 19. The bathtub model of atmospheric CO_2.

Ethanol fuel [P1 Q21]: pros	Cons
Renewable: crops can be regrown. Plants absorb CO_2 as they grow, released again when the ethanol burns — so it is **close to carbon-neutral**.	Needs lots of water, land and fertilizer; can raise food prices. Farm machines and transport run on fossil fuels — so it is **not exactly** carbon-neutral (your own con shows why).

CO_2-free electricity [P1 Q22]: **solar, wind, geothermal** — √ correct (hydroelectric is another). Your point that solar is promising with batteries for night is a good one.

Part 9 · Mauna Loa & the Keeling Curve

Mauna Loa is a still-active volcano in Hawaii; its observatory, about **3,400 m** high, samples large, well-mixed air masses [P3]. **Charles David Keeling** began measuring CO_2 there in **March 1958** at about **313 ppm** (Scripps first reading; ppm = parts per million: CO_2 molecules per million air molecules); the graph of his data is the **Keeling Curve**. **May 2026:** record peak, about **432 ppm** (Scripps; NOAA). **Sept 2026:** about **426 ppm** (NOAA, preliminary). Pre-industrial (before about 1800): about **280 ppm** (ice cores).

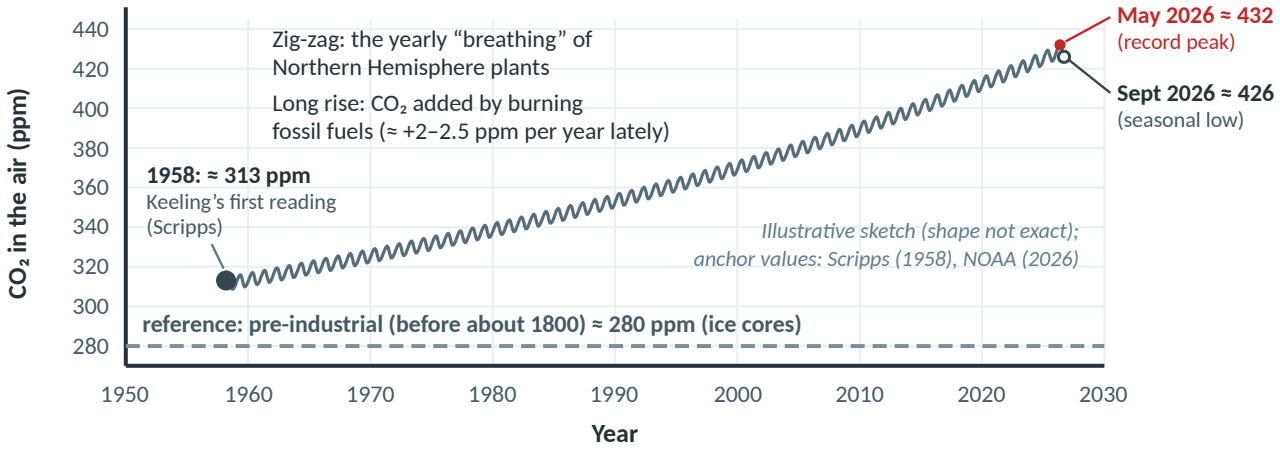


Figure 20. The Keeling Curve (illustrative sketch, real anchor values). CO_2 rises even though it drops every summer, because each year more is added (mostly by burning fossil fuels) than removed — recently about +2–2.5 ppm per year on average.

Why the yearly zig-zag? The running total

The CO_2 level is a **running total — like water in a bathtub**. In fall and winter, leaves drop and photosynthesis slows, but respiration and decay (microbes breaking down fallen leaves) keep adding CO_2, so the level **keeps rising** until it peaks in **May**, just as leaves come back. Months of strong summer photosynthesis then pull it down to the low in **Sept/Oct**. The swing comes mostly from the huge forests of the **Northern Hemisphere**. The Southern Hemisphere has the opposite pattern, and a weaker one, because it has **less land** and so fewer land plants [P3].

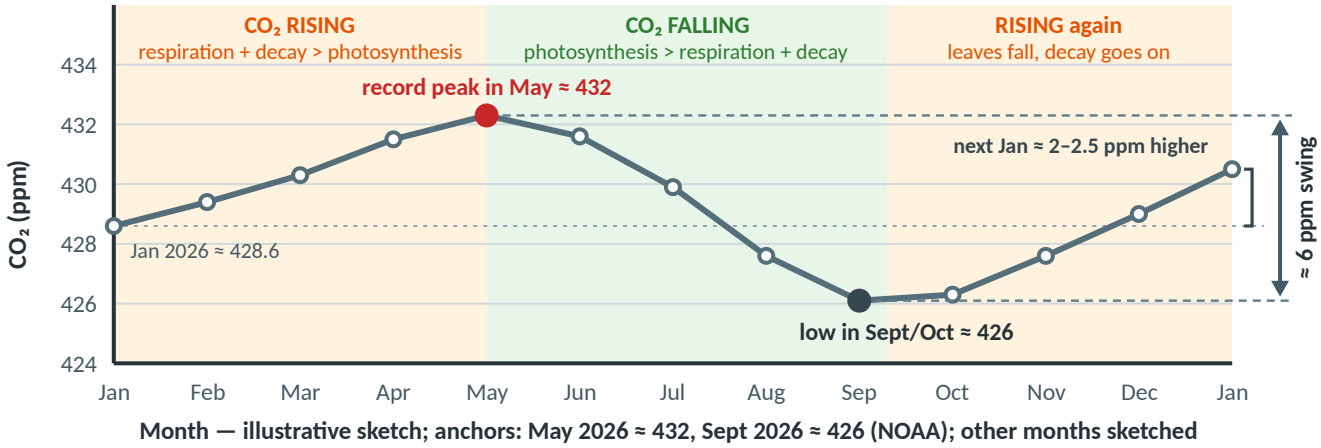


Figure 21. One year at Mauna Loa (illustrative): CO_2 climbs to a May peak and falls to a Sept/Oct low — a swing of about 6 ppm.

Two fix-its from your Mauna Loa notes [P3 p.1]

√ FIX-IT FROM YOUR WORKSHEET — "Photosynthesis is a delayed reaction"

Partly right: the **observation** is correct (more CO_2 in spring than fall; peak in May, low in Sept/Oct), and the *level* does lag behind the seasons. But photosynthesis itself starts within seconds of light. The lag happens because the level is a **running total**: it keeps climbing until photosynthesis removes more than respiration and decay add.

√ FIX-IT FROM YOUR WORKSHEET — "Cellular respiration... has a negligible effect on CO_2"

Likely a note-taking slip. Respiration (plants, animals and especially decomposers) releases a **huge** amount of CO_2 — about as much as photosynthesis removes each year (the balance in your own [P1 Q18]) — and it pushes CO_2 up every fall and winter. The lesson probably meant: respiration goes on **all year**, so the big *season-to-season change* comes mostly from photosynthesis switching up and down.

§ WHERE THIS COMES FROM — Other notes, polished [P3]

- "Regular" vs. "rampant" CO_2 (class terms) = CO_2 from **natural** sources vs. **human-caused** CO_2 — the same molecule, a different source. And respiration **breaks down** glucose (with O_2) to make ATP.
- CO_2 began rising with the **Industrial Revolution (1800s)**, from a pre-industrial level (before about 1800) of about 280 ppm; the rise **sped up sharply after WWII**.
- √ Right in your notes: extra CO_2 from burning fossil fuels builds up faster than Earth can absorb it, and more CO_2 traps more heat. (Deforestation is a second human source [P1 Q20].)

Part 10 · Vocabulary

Term	Plain-English meaning	Example
Photosynthesis	Producers use light energy to make glucose + O_2 from CO_2 + water.	leaves on a sunny day

Term	Plain-English meaning	Example
Cellular respiration	Cells break down glucose with O_2 to release its energy as ATP (+ CO_2 and water).	muscle cells; plants, 24/7
Chloroplast	The organelle (cell part) where photosynthesis happens.	green ovals in leaf cells
Chlorophyll	Green pigment that absorbs light (mostly red and blue/violet).	why leaves look green
Mitochondrion (pl. mitochondria)	The organelle where cellular respiration makes ATP.	in plant AND animal cells
Glucose ($C_6H_{12}O_6$)	A sugar that stores chemical energy.	stored as starch in seeds
ATP	The cell's small, usable packets of energy ("energy currency").	powers muscle movement
Reactant / product	What goes into a reaction / what comes out of it.	CO_2 in / O_2 out
Light / chemical energy	Energy carried by light / stored in the bonds of molecules.	photosynthesis: light → chemical
Stomata	Tiny leaf pores that let CO_2 in and O_2 out.	mostly under the leaf
Producer / consumer / decomposer	Makes its own food / eats others / breaks down dead matter.	grass / rabbit / mushroom
Interdependence	Living things depend on each other.	fish give CO_2; plants give O_2
Dissolved oxygen (DO)	O_2 gas dissolved in water, measured in mg/L.	the simulation's DO reading
DO difference (net)	DO end − DO start = O_2 made − O_2 used.	+0.4037 at 100 % light
O_2 production (gross)	Total O_2 made by photosynthesis = DO difference + 0.1939.	0.5976 at 100 % light
Independent / dependent variable	What you change on purpose / what you measure.	light % / O_2 production
Controlled variable	Kept the same so the test is fair.	25 °C, white light, 0 fish
Control (baseline)	A comparison run with the factor removed.	Part II dark run
Limiting factor	The factor in shortest supply that holds the rate back.	light, 40–100 % in Part III
Break-even point	Light level where O_2 made = O_2 used (light compensation point).	near 60 % in your data
Enzyme / denature	Protein that speeds up a reaction / loses its shape in high heat.	most plant enzymes: 40 °C+
Optimum temperature	The temperature at which a process works best.	best tested ≈ 20 °C
Productivity	The rate at which producers make organic compounds (food).	estimated from DO
Carbon cycle	The path carbon atoms take between the air, living things and underground stores.	Model 2 [P1]
Combustion / fossil fuels	Burning (makes CO_2) / coal, oil, gas formed over millions of years.	cars, power plants
Carbon-neutral	Adds no more CO_2 to the air than it removes overall.	ethanol: close, not exactly
Equilibrium (balance)	Amount in ≈ amount out, so the total stays steady.	natural carbon cycle [P1 Q18]
ppm / Keeling Curve	CO_2 molecules per million air molecules / the Mauna Loa CO_2 graph.	≈ 432 ppm in May 2026

Part 11 · Quick self-check

Cover the answers (or use the flip cards in the HTML version). If you miss one, reread the part in brackets.

1. Write the photosynthesis equation. (Part 2)
Answer: $6CO_2 + 6H_2O$ + light energy → $C_6H_{12}O_6 + 6O_2$

2. Write the cellular respiration equation. (Part 3)
Answer: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy (ATP)

3. Which energy goes IN to photosynthesis, and which comes OUT of respiration? (Part 4)
Answer: Light energy goes in; respiration releases energy as ATP (usable energy), plus some heat.

4. Do plants do cellular respiration? When? (Part 3)
Answer: Yes — in their mitochondria, all the time, in light and dark.

5. How is breathing different from cellular respiration? (Part 3)
Answer: Breathing moves gases in and out of the lungs; cellular respiration is a chemical reaction in mitochondria that makes ATP.

6. Why must sunlight be constantly available? (Part 1)
Answer: Energy is not recycled — it flows through living things and leaves as heat [P1 Q4].

7. What does the DO difference measure? How do you get O_2 production from it? (Part 5)
Answer: The net change: O_2 made − O_2 used. O_2 production = DO difference − (−0.1939) = DO difference + 0.1939.

8. O_2 production at 100 % and at 20 % light? (Part 5)
Answer: 0.5976 and 0.0011 mg/L per hour.

9. What happened near 60 % light? (Part 5)
Answer: DO difference ≈ 0 (+0.0046): break-even, O_2 made ≈ O_2 used (one trial).

10. What was the limiting factor in Part III, and how do you know? (Parts 5–6)
Answer: Light: O_2 production rose ≈0.20 mg/L per hour per +20 % of light from 40 % up, with no leveling off.

11. Name the independent, dependent and controlled variables in Part III. (Part 5)
Answer: IV: light intensity. DV: O_2 production (mg/L per hour). Controlled: 0 fish, 1 plant, 25 °C, white light, 60 min.

12. Why is the Part II dark run the control? (Part 5)
Answer: No light → no photosynthesis, so it measures respiration alone: the baseline (−0.1939 mg/L per hour).

13. Which colors does chlorophyll absorb best? (Part 6)
Answer: Red and blue/violet; green is reflected (why leaves look green).

14. Why is "enzymes denature just above 20 °C" wrong? (Part 6)
Answer: Above the optimum enzymes just work less efficiently and respiration speeds up; real denaturing needs about 40 °C or more.

15. Which Model 2 arrows add CO_2 to the air, and which removes it? (Part 7)
Answer: A, B, C (respiration) and combustion add CO_2; D (photosynthesis) removes it.

16. Why does CO_2 keep rising if plants remove it every summer? (Parts 8–9)
Answer: Each year more is added (mostly fossil fuels) than removed — recently about +2–2.5 ppm per year.

17. Why does Mauna Loa CO_2 peak in May and hit a low in Sept/Oct? (Part 9)
Answer: All fall and winter respiration and decay add more than photosynthesis removes, so the running total climbs until leaves return in May; summer photosynthesis then draws it down until Sept/Oct.